Project Name: “SportsBetPro: Enhancing Predictive Modeling for Sports Betting"

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Brief Description:
"SportsBetPro" focuses on leveraging advanced regression and classification models to predict outcomes in sports betting, specifically targeting spreads (point differential assigned by oddsmakers to balance the perceived strength of two opposing teams), over/unders (prediction of the combined score of both teams compared to a set line) and moneyline results (straightforward method of betting on the outcome of a sports event, where betters pick which team they believe will win the game outright) across the National Basketball Association (NBA).

Problem Statement:
Many sports betting enthusiasts struggle with accurately predicting game outcomes. Existing predictive models often lack precision or overlook crucial factors, leading to inconsistent results. Our models strive to offer a way to efficiently make calculated betting decisions.

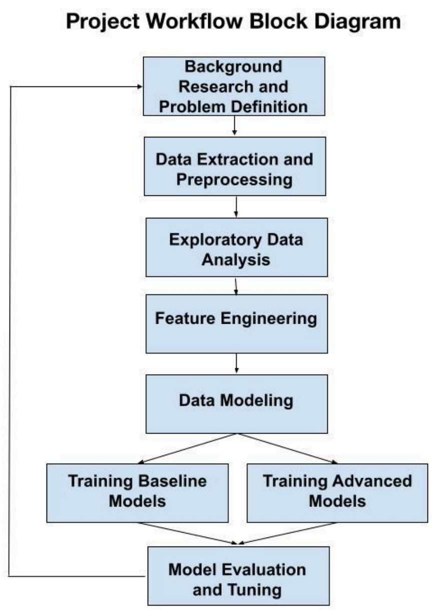
Objective:
The objective of "SportsBetPro" is to develop and evaluate models capable of predicting spreads, over/unders, and moneyline outcomes in sports betting with low error and positive profit. Our aim is to achieve an accuracy rate of at least 55% on our predictions, providing users with reliable tools for informed decision-making in sports betting scenarios.

Approach/ Methodology:
To achieve the objectives of our project, we employed a comprehensive approach to train, develop, and evaluate our models. Our methodology encompassed the following key steps:

1. Data Collection: First, our team searched publicly available datasets containing historical betting odds, player statistics, and game outcomes across multiple seasons of the NBA.
2. Exploratory Data Analysis: Our team explored the relationship between features to gain insights into the structure and characteristics of the datasets, identifying key features and distributions of the data relevant to our predictive modeling objectives.
3. Model Preprocessing: Feature engineering techniques were applied to preprocess the data and extract meaningful features that could enhance the predictive performance of our models. In addition, we explored dimensionality reduction techniques such as feature selection and Principal Component Analysis (PCA) to optimize model performance.
4. Model Training: We identified each research question as a regression or classification problem, and trained several models for each of the given tasks based on the features generated in the preprocessing stage. We split our feature set into an 80-20 train-validation split to train each model. The most recent season in our dataset (2022) was reserved as a testing set.

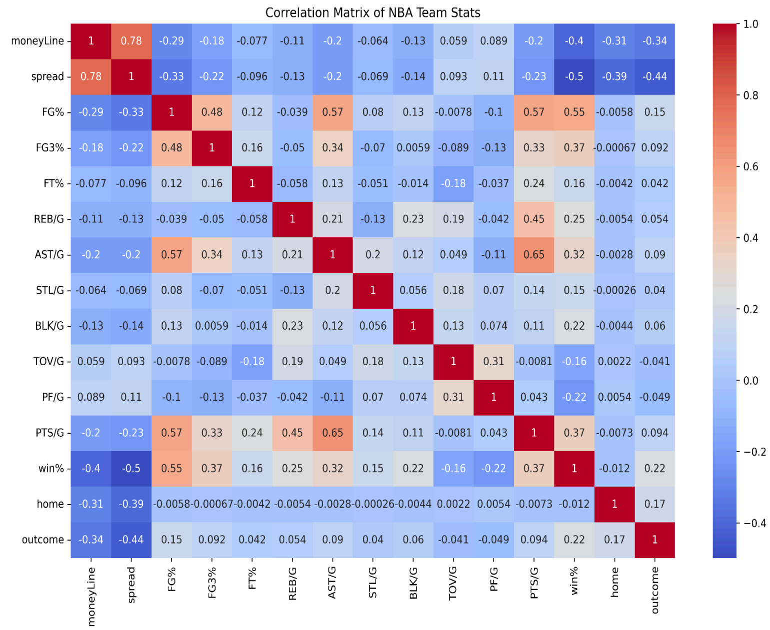
5. Model Evaluation: After training each model, we evaluated and optimized each model based on common evaluation metrics for the training, validation, and testing set.
6. Impact Analysis: Finally, we analyzed the potential impact of our models by comparing the projected outcomes with actual results and quantifying the expected financial gains or losses. This step provided valuable insights into the tangible utility of our models.

Block Diagram:

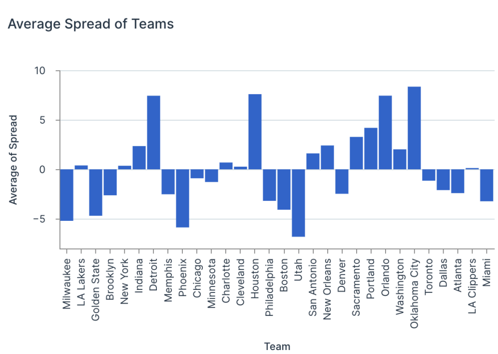


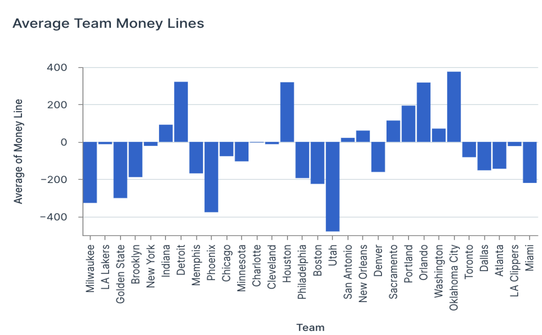
Datasets:
The "SportsBetPro" project leverages several key datasets, obtained from Kaggle, to develop and evaluate regression and classification models for predicting spreads, over/unders, and moneyline outcomes in sports betting across the NBA. The first dataset utilized is the NBA Player Stats dataset, which contains statistics and performance metrics for NBA players from 1950-2022, offering insights into team offensive and defensive performance. This data is crucial for creating aggregated team statistics and understanding the contributing factors for predictive modeling. Additionally, the project relies on the NBA Odds Dataset, providing information on NBA betting lines, final game scores, and original lines set by sportsbooks like Vegas. To effectively handle the temporal aspect of the data, each season was split into its own dataset. The team implemented a strategy where they created a dictionary of tuples, each containing a separate dataframe for each season and team. This approach facilitated the organization and analysis of the data on a season-by-season basis.

Images:

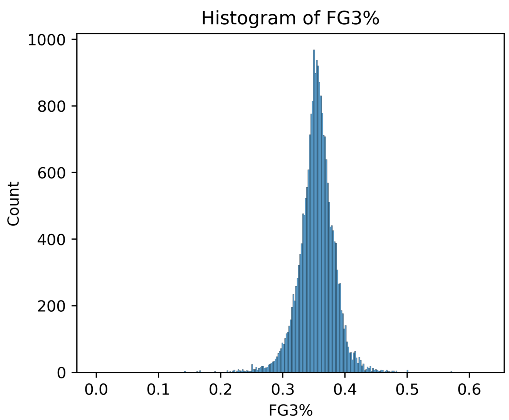


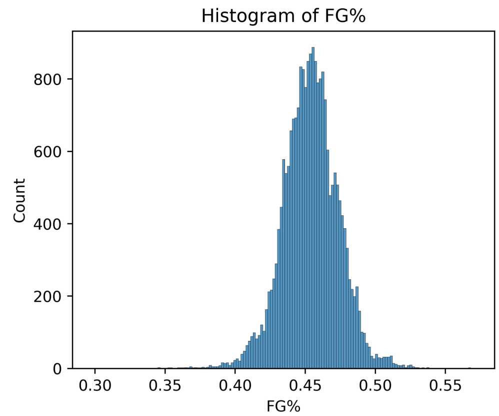
This correlation heatmap serves as a good sanity check for our data. We see that the highest correlations are in line with what we would expect from sports logic. Field goal percentage is strongly positively correlated with winning percentage and points per game. This makes sense because teams are more likely to win if they choose wise shots and consistently make shots from the field.





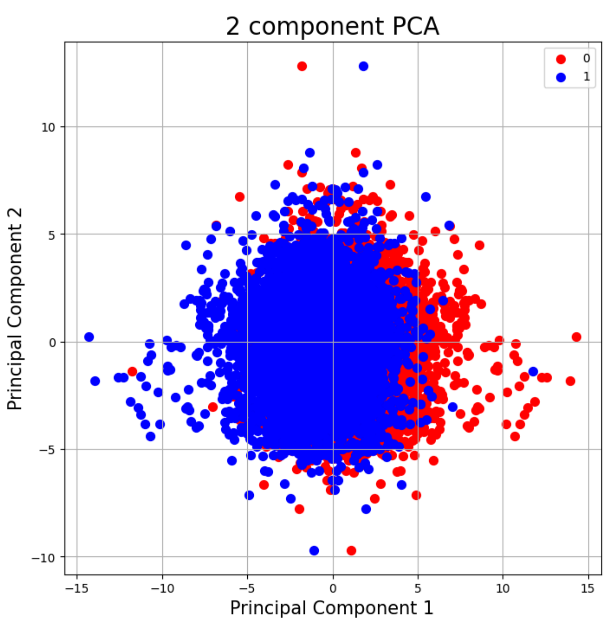
One key takeaway we can get from the money line and spread data is that they mirror each other well, highlighting their strong correlation from the heatmap above.

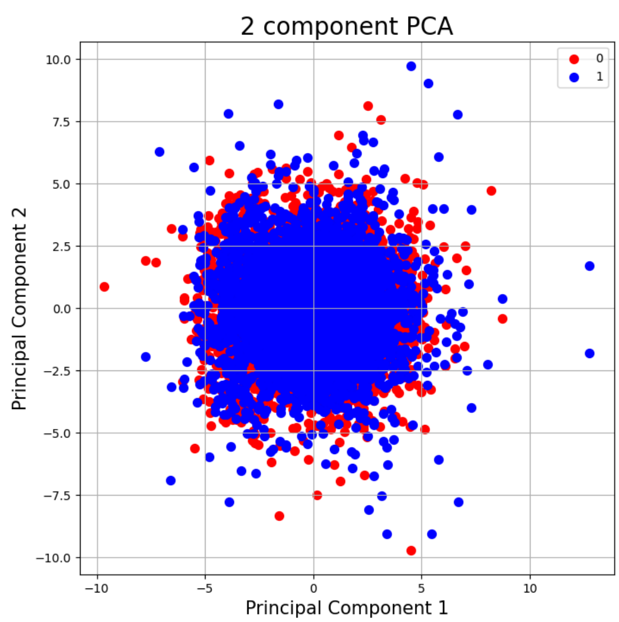




We included the field goal percentage histograms because of the high kurtosis, showing how important proportional statistics are towards game success. A 5% difference in field goal percentage or 3-point field goal percentage may be the difference between victory and defeat, which is what we often see in many basketball games.

Additionally, in an attempt to reduce the dimensionality of our features, we performed Principal Component Analysis. However, when plotting the principal components for each of the two classification problems, we did not see any separability among features, and chose to keep the original feature set (win/loss on the left, over/under on the right)





Other attempts to reduce the number of features were attempted (such as picking the most “important” features from models) but all such attempts resulted in a significant reduction in expected monetary profit.

What is Considered Success/Failure?

Success:

1. Proportion of Correct Bets: A successful model would correctly predict more than 55% of its bets. This threshold was chosen based on historical data and industry standards, as sports bettors typically see this figure as the minimum to make a profit.
2. Comparison to Vegas: Another standard of success compares our model's performance to that of professional sportsbooks. These books typically achieve around 80% betting accuracy, providing a measure to compare our model against professional standards.
3. Positive Profit: Ultimately, the primary objective of our project is to enable sports bettors to make informed decisions that result in positive financial outcomes. Therefore, achieving profitability through our model predictions constitutes a significant measure of success. Positive profit indicates that our models are effectively identifying value bets and outperforming random chance or naive strategies.

Failure:

1. Proportion of Correct Bets: Falling short of the 55% threshold signifies a failure to meet the minimum requirement for effective predictive modeling in sports betting. Models with accuracy below this threshold may not provide sufficient value or reliability to users, leading to suboptimal betting decisions and potentially negative financial outcomes.
2. Negative Profit: Similarly, experiencing negative profitability as a result of our model predictions indicates a failure to enhance betting strategies. Negative profit suggests that our models are not generating accurate predictions, resulting in financial losses.

Evaluation Parameters:

In evaluating the performance of our predictive models, we employed a range of evaluation metrics to assess their accuracy, robustness, and effectiveness.

1. Accuracy: Measures the exact proportion of bets that were predicted correctly. Mostly used in our classification models, but also useful when evaluating the betting performance of regression models. A higher accuracy means a higher proportion of bets were correctly predicted.
2. Precision: A measure of how many true positives are predicted, relative to the number of false positives. A higher precision means there are fewer false positives.
3. Recall: A measure of how many true negatives are predicted, relative to the number of false negatives. A higher recall means there are fewer false negatives.
4. F1-Score: A composite metric that considers both precision and recall to provide a balanced measure of a model's performance.
5. Root Mean Squared Error: A loss metric measuring the average distance between a predicted value and an actual value. Used exclusively for our regression tasks. A lower RMSE means a closer prediction.
6. Expected Profit: We calculated how much money would be gained or lost from betting $100 on every result predicted by one of our models.

Experiments:

- Win/Loss Classification - for this problem, we employed several binary classification models: logistic regression, Gaussian Naive Bayes, Random Forest, Support Vector Machine (SVM), AdaBoostClassifier, and a Neural Network. Two deep learning models were also generated for this task in Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU). Our models aim to predict whether a team will win or lose a game outright, with a class of 1 assigned to wins and 0 assigned to losses.
- Over/Under Regression - we first attempted a regression approach to predict the total score between the two teams in a game. The models we employed for this task were linear regression, random forest regressor, and a multi-layer perceptron.
- Over/Under Classification - after our regression models, we attempted a binary classification approach towards over/under prediction, this time striving to predict whether or not the teams' total score would exceed or fall short of the line set by Las Vegas. The models we employed include: Logistic Regression, Gaussian Naive Bayes, Random Forest, and AdaBoostClassifier.
- Spread Regression - our final modeling task was to predict the score difference between the two teams that played in the game. For this, we employed linear regression, a random forest regressor, and a multi-layer perceptron.

Results:

Win/Loss Classification (Testing Set)

Model	**Precision**	**Recall**	**F1-score**	**Accuracy**	**Expected Earnings**
Logistic Regression	0.67716	0.68200	0.67957	0.67843	1416.83
Gaussian Naive Bayes	0.65927	0.66871	0.66396	0.66155	-3034.67
Random Forest	0.67662	0.69530	0.68583	0.68149	3641.07
Support Vector Machine	0.68215	0.66053	0.67112	0.67638	645.80
AdaBoost	0.68054	0.66871	0.67458	0.67740	1538.35
Custom Neural Network	0.67587	0.67030	0.68109	0.67587	458.85
LSTM Model	0.59618	0.44518	0.50973	0.5711	-2635.67
GRU Model	0.58620	0.63377	0.60906	0.5925	2104.20

Over/Under Regression (Testing Set)

Model	**RMSE**	**Proportion Correct**	**Expected Earnings**
Linear Regression	19.48	0.45910	-22180
Random Forest	19.40	0.46575	-19450
MLP	19.59	0.47035	-17560

Over/Under Classification (Testing Set)

Model	**Precision**	**Recall**	**F1-score**	**Accuracy**	**Expected Earnings**
Logistic Regression	0.49298	0.32377	0.39085	0.49642	-11250
Gaussian Naive Bayes	0.50463	0.55738	0.52970	0.50163	-7260
Random Forest	0.54167	0.02664	0.05078	0.50307	-8520
AdaBoost	0.46974	0.36578	0.41129	0.47751	-19020

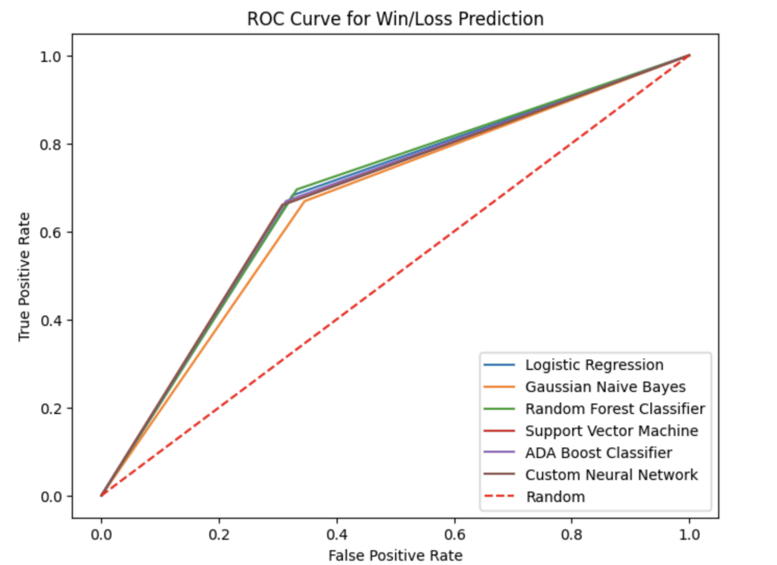
Spread Regression (Testing Set)

Model	**RMSE**	**Proportion Correct**	**Expected Earnings**
Linear Regression	13.60	0.51125	-1420
Random Forest	13.65	0.48160	-13600
MLP	13.75	0.48364	-12760

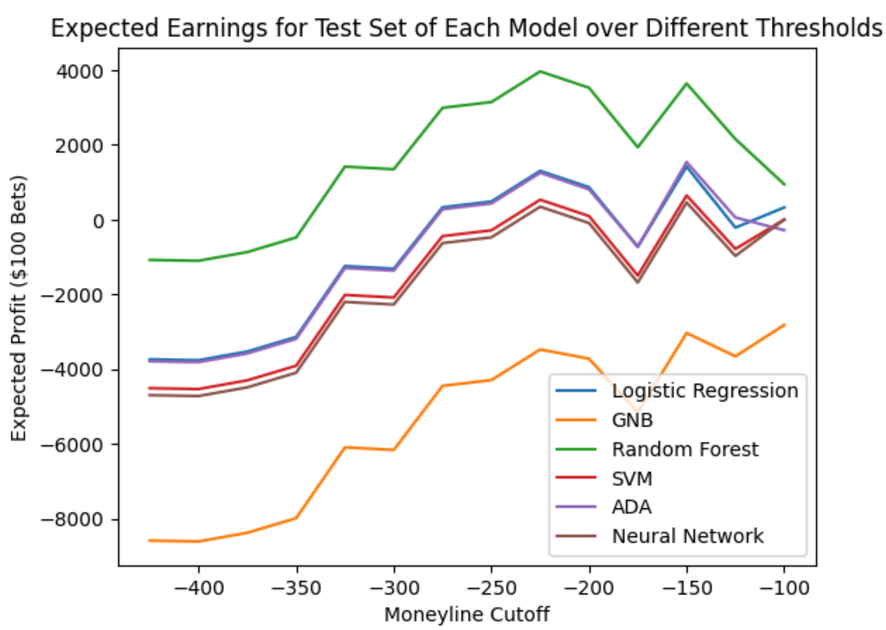
Tests/Graphs/Discussions:

Win/Loss Prediction:

Of all our research questions, our models performed the best for win/loss classification by a wide margin. Below are the ROC curves for all models. Overall, each model has a similar trend when it comes to the relationship between False Positive Rate and True Positive Rate.



This was also the only research question where our models consistently made a profit. One additional aspect to consider when predicting wins and losses is that professional sportsbooks often balance how much money is actually won when a certain team wins. This is to balance out any team being a perceived favorite. As a result, we analyzed the various thresholds (known as money lines) and saw how our expected winnings would benefit:



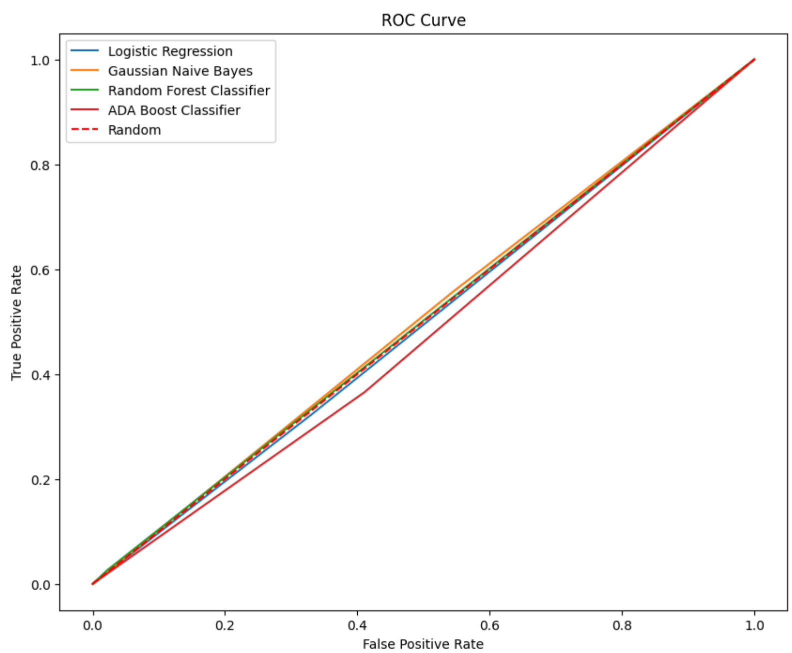
Most models seemed to peak at a threshold of –150, so we calculated the expected winnings above based on that number.

Over/Under Regression:

This research question was a strange case. On one hand, the proportion of bets in the test set that were correctly predicted was the lowest of all the models. On the other hand, this means that if we were to bet the opposite of our model, the results would be fairly promising. Going against our model's predictions yields expected profits of $7280, $4420, and $2160 for linear regression, random forest, and MLP regressors respectively. As a result, even though the performance of the models is poor, they are still a useful tool to maximize betting outcomes.

Over/Under Classification:

The over/under classification models all have an accuracy hovering around 0.5. This means that our models are not doing a great job of differentiating which games will be high or low scoring, and therefore often have to guess the result. The ROC curve of the models below further illustrates this issue:



The likely reason for this poor performance is the features that were available to us. Our features often illustrate how certain teams are able to perform relative to one another, which is why we are able to predict who will win or lose a certain game. However, these features aren't as illustrative when it comes to identifying whether the total score would reach above a certain threshold. Another contributing factor is professional sportsbooks adjusting the over/under line to be different for each game. As a result, it is more difficult by design to predict this problem as opposed to the win/loss problem.

Spread Regression:

Similar to over/under, the spread regression models often hit around 50% of their predicted bets. Once again, this is likely due to the fact that our features are well-suited to predict who would win or lose a game, but are less able to predict the final point differential between teams. However, something interesting to note is that the predictions from the linear regression model are only expected to lose money because most sportsbooks take about 10% of all earnings (which we accounted for). If you take away this 10%, then the linear regression predictions are able to make a profit of $7800.

Comparison:

Of all our research questions, the most successful was the win/loss prediction. Once again, this was likely due to the structure of our features. Common team level basketball stats such as points per game, rebounds per game, blocks per game, etc. are all very helpful in determining which team is better than the other, but aren't as helpful when trying to predict how many points are scored in total (over/under), or the final score differential (spread). The combination of these features with the seamless fit into a binary classification task framework makes the win/loss prediction task perform the best. Our metrics as shown in the figures above support this assumption, with our F1 score, precision, recall, and accuracy all consistently being very high all across the entire board on our test set. They perform significantly higher than the classification task for over-unders, which don't perform much higher than the baseline with expected 50%. With our regression models we see that about 50% of the spreads from the models were within the actual results of the game, but the expected earnings from the predictions were all negative, which isn't the best indicator for a model in this context.

Overall, we wanted a predictive model that performs better than the 50% accuracy rate that we would get with no sports knowledge, and the regression models nor the over-under classification models provided that. The win/loss classification models provided us with the best results for this case, giving us the most confidence in those models.

Constraints:

To simplify and standardize our study, several constraints were placed on our data:

1. Limited Timeline: Because sports strategies and skill levels are always changing, our team limited our study to only include games that took place in the 2009-2010 NBA season or later.
2. Absence of Individual Player Data: To maintain a game-by-game granularity, we leveraged aggregated team statistics to model game outcomes and opted not to use player-specific information, such as injury histories, performance metrics, and playing styles. This limits the depth of our data, and future iterations may explore how individual player data may contribute to our predictions.
3. Sportsbook Lines: Different professional sportsbooks often set different betting lines. To standardize our analysis, we solely focused on the betting lines present in our dataset. In addition, when evaluating our expected winnings, we assumed a $110 loss for wrong bets and a $100 gain for correct bets.

Standards:

Software Packages Selection:

- Pandas (2.1.4) for data manipulation and preprocessing.
- NumPy (1.23.4) for numerical computing and array operations.
- Scikit-learn (1.1.2) for machine learning algorithms and model evaluation.
- TensorFlow (2.10.0) and Keras (2.10.0) for deep learning and neural network architectures.
- Matplotlib (3.6.0) and Seaborn (0.13.2) for data visualization and exploration.
- Deepnote Notebook for interactive development and documentation.

Limitations of the Study:

1. Use of Aggregated Team Data: One limitation of our study is the reliance on aggregated team data rather than individual player data. While team statistics provide valuable insights into overall performance, they may overlook the impact of individual player dynamics, such as star players' injuries or rest days.
2. Unpredictability of Extraneous Factors: Certain factors such as in-game injuries, last-minute changes, etc., can affect the outcome of a game and cannot be predicted with a model. While our models aim to capture as many relevant variables as possible, the inherent unpredictability of these factors introduces uncertainty into our predictions.
3. Temporal Variability: The inclusion of data points from different seasons introduces temporal variability, wherein the importance and relevance of certain statistics may evolve over time. For example, changes in game strategies or rule modifications, such as the increased emphasis on three-point shooting in recent seasons, can alter the significance of certain performance metrics.
4. Specific Matchup Dynamics: Divisional matchups, playoff-clinching scenarios, and other situational dynamics are not explicitly accounted for in our models. These factors can influence team strategies and performance.

Future Work:

1. More Specialized Features: Currently, our features are very good at determining which team is better than the other, but are unable to help predict our other research questions. Adding more features that are more appropriate for our other research questions and then specializing our feature matrices will help provide a boost to our performance.
2. Incorporating Most Recent Data: One avenue for future work involves incorporating the most recent NBA seasons' data into our predictive modeling framework. By leveraging up-to-date statistics and performance metrics, we can capture the latest trends, patterns, and dynamics in team and player performance.
3. Enhanced Model Tuning: Continuously fine-tuning and optimizing our models represent another crucial area for future work. This includes exploring advanced feature engineering techniques, refining model hyperparameters, and experimenting with alternative algorithms to improve predictive performance and robustness.

4. Incorporating Dynamic Factors: Future iterations of our predictive models could aim to incorporate dynamic factors such as injuries, rest days, back-to-back games, trades, roster moves, and coaching changes.
5. Advanced Statistical Techniques: Exploring advanced statistical techniques and machine learning algorithms, such as time-series analysis, Bayesian methods, reinforcement learning, and deep learning architectures, offers promising avenues for future research. These approaches can capture complex temporal dependencies, nonlinear relationships, and latent patterns in sports betting data, potentially leading to more sophisticated and accurate predictive models.
6. Evaluation of Profitability: Conducting comprehensive evaluations of model profitability, including backtesting against historical data and real-world betting scenarios, could help us better understand the real-world impact of our models.

Human Context and Ethics:
In the context of sports betting and predictive modeling, considerations of fairness are paramount to ensure equitable treatment and mitigate potential biases or disparities. Our analysis acknowledges several factors that may influence the fairness of our predictive models.

1. Influence of Market Size: Small market teams face inherent disadvantages such as limited resources, smaller fan bases, and reduced media exposure, impacting their ability to attract top talent and compete for championships. Conversely, larger market teams enjoy advantages like higher salary caps and better facilities, giving them a competitive edge. In sports betting, these disparities may introduce biases, favoring larger market teams and influencing predictive models. Recognizing and addressing these biases is essential to ensure fair treatment and accuracy in our predictions across all teams.
2. Addressing Bias and Promoting Fairness: To promote fairness in predictive modeling, we must adopt rigorous methodologies that account for team performance, player talent, coaching strategies, and game dynamics. By incorporating diverse data sources and advanced statistical techniques, we aim to develop robust, accurate, and unbiased models. Ongoing monitoring and refinement based on real-world feedback can help identify and address emerging biases, ensuring equitable treatment in sports betting predictions.

Overall, promoting fairness in sports betting predictive modeling requires a nuanced understanding of the dynamics and challenges inherent in the sports industry, including the influence of market size on team performance and the potential biases that may arise as a result. By actively addressing these factors and striving to develop fair and transparent predictive models, we can support responsible and equitable decision-making in sports betting for all stakeholders involved.